

☰ Hide menu

Introduction

Introduction to Linear Models

Reading the Data

Implementing Linear Models for Image Classification

Neural Networks and Deep Neural Networks for Image Classification

Deep Neural Networks with Dropout and Batch Normalization

Introduction

Convolutional Neural Networks

Understanding Convolutions

CNN Model Parameters

Working with Pooling Layers

Implementing CNNs on Vertex AI by using a pre-built TF container

Introduction

Preprocessing the image data

Model parameters and the data scarcity problem

Data Augmentation

Transfer Learning

🎥 Video: Transfer Learning
6 min

🎥 Video: Lab Introduction - Classifying Images with Transfer Learning
11 sec

📎 Graded App Item: Lab: Classifying Images with Transfer Learning
1h 30m

📖 Reading: Readings
30 min

📄 Quiz: Quiz

Summary

Quiz

✔ Submit your assignment

Due Feb 11, 11:59 PM IST

✔ Receive grade

To Pass 80% or higher

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✔ Congratulations! You passed!

Grade received 100%

Latest Submission Grade 100%

To pass 80% or higher

Go to next item

		Try again
1.	1 / 1 point	
What is data augmentation?		
☐ Data augmentation is a technique where randomly selected neurons are ignored during training. ☐ Data augmentation is the grouping together of resources for the purposes of maximizing advantage or minimizing risk to the users. ☒ Data augmentation is a set of techniques that enhance the size and quality of training datasets with the goal of creating more accurate ML models that generalize better. ☐ Data augmentation is the amount of pixels added to an image when it is being processed by the kernel of a CNN.		Your grade 100% View Feedback We keep your highest score
2.	1 / 1 point	
How does preprocessing help to improve the quality of the image?		
☐ Preprocessing increases unwanted distortions and enhances the required features that are essential for the application. ☒ Preprocessing suppresses unwanted distortions and enhances the required features that are essential for the application. ☐ Preprocessing improves the image data by inducing missing values, noisy data, and other inconsistencies before executing it to the algorithm. ☐ Preprocessing increases unwanted noise and controls the quality of the image.		
☒ Correct Correct. Before raw images can be fed into an image model, they usually have to be preprocessed. These preprocessing operations can include resizing, converting between color spaces, cropping, flipping, rotating and transposing for shape transformation or image adjustments, segmentation, and compression for quality enhancement.		
3.	1 / 1 point	
How does transfer learning deal with the data scarcity problem?		
☐ Transfer learning takes the pre-existing samples and changes them in some way to create new samples and also increase the number of training samples, and is typically used with image data. ☐ Transfer learning increases the number of parameters to deal with data scarcity. ☒ Transfer learning transfers knowledge across tasks so, instead of creating more data, transfer learning decreases the need for data by initializing the parameters with better values. ☐ Transfer learning boosts the need for data by initializing the parameters with better values.		
☒ Correct Correct. Transfer learning uses knowledge acquired for one task to solve related tasks.		
4.	1 / 1 point	
What is negative transfer learning in computer vision?		
☐ When knowledge is transferred from a less related source, the target performance is not degraded. ☒ When knowledge is transferred from a less related source, the target performance might be degraded. ☐ When labeled data for a specific target task is abundant, target performance is not degraded. ☐ When labeled data for a specific target task is scarce, the target performance is enhanced.		
☒ Correct Correct. When labeled data is scarce for a specific target task, the target performance is affected.		
5.	1 / 1 point	
What is the proportion of the number of parameters in the entire network while computing?		
☐ A small number of parameters comes from the dense layers at the end, and the convolutional layers also contain far fewer parameters. ☐ A large number of parameters comes from the dense layers at the end, along with the convolutional layers. ☐ The number of parameters from dense layers and convolutional layers has the same proportion. ☒ A large number of parameters comes from the dense layers at the end, and the convolutional layers contain far fewer parameters.		
☒ Correct Correct. To compute the number of parameters in a convolutional layer, multiply the number of parameters per filter by the number of filters and divide by the stride, and then finally add bias terms for each filter.		

